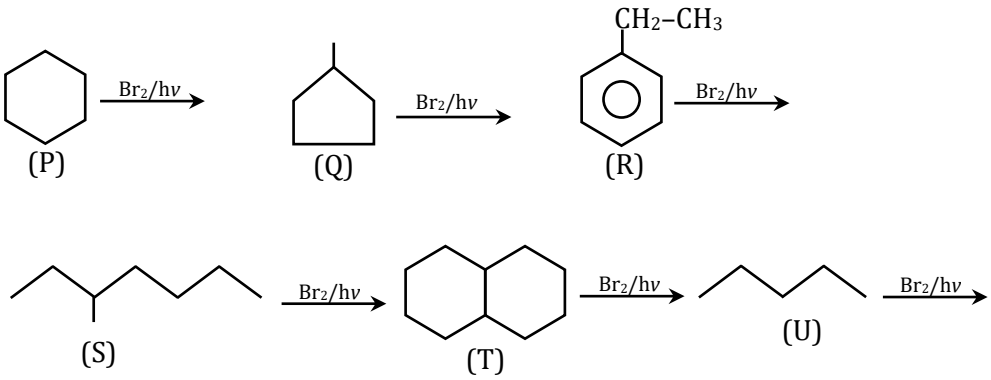
RACE
Regular Analysis Through Continuous Exercise

Reaction Mechanism : RACE-01

Carbocation, Freeradical substitutaion, Elimination Reactions

1. Among the following free radical bromination reactions, select those in which 2° halide is the major product.



(A) P, Q, R, S (B) P, R, U (C) P, R, S, T (D) P, Q, R, S, T

CRM132

2. Select the chain propagation steps in the free-radical chlorination of methane.

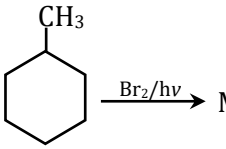
(1) $Cl_2 \longrightarrow 2Cl^{\bullet}$ (2) $Cl^{\bullet} + CH_4 \longrightarrow CH_3Cl + H^{\bullet}$

(3) $Cl^{\bullet} + CH_4 \longrightarrow HCl + \dot{C}H_3$ (4) $H^{\bullet} + Cl_2 \longrightarrow HCl + Cl^{\bullet}$

(5) $CH_3^{\bullet} + Cl_2 \longrightarrow CH_3Cl + Cl^{\bullet}$

(A) 2, 3, 5 (B) 1, 3, 6 (C) 3, 5 (D) 2, 3, 4

CRM133

3. 

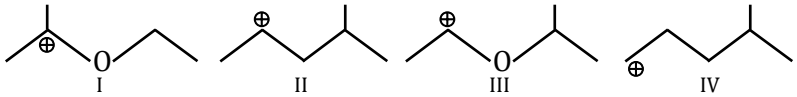
Monobromo derivatives

The number of possible monobromo products is (Excluding stereoisomerism)

(A) 4 (B) 5 (C) 8 (D) 10

CRM134

4. The correct stability order of the following species is :

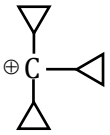


(A) II > IV > I > III (B) I > II > III > IV (C) II > I > IV > III (D) I > III > II > IV

CRM135

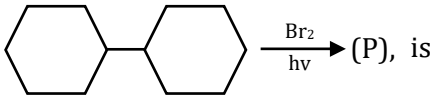
For More Join: @IITJEE_Advanced

5. The more stable carbocation from the following is :

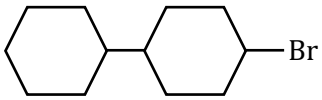
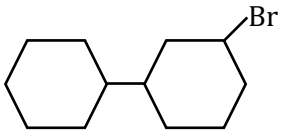
(A) $(CH_3)_3\overset{\oplus}{C}$ (B) $Ph\overset{\oplus}{C}H_2$ (C)  (D) All of these

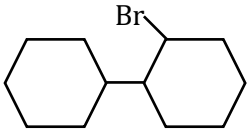
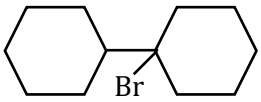
CRM136

6. Major product (P) for the reaction



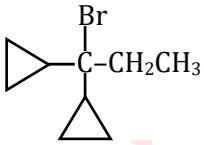
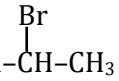
(P), is

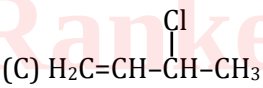
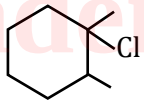
(A)  (B) 

(C)  (D) 

CRM137

7. Which of the following compounds will given E1 reaction?

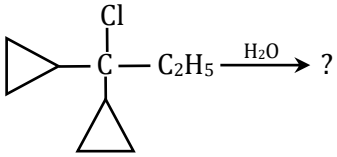
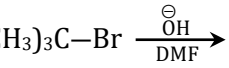
(A)  (B) 

(C)  (D) 

Rankers Academy JEE

CRM138

8. Which of the following will undergo E1elimination?

(A)  (B) 

(C) Both A and B (D) None of these

CRM139

ANSWER KEY

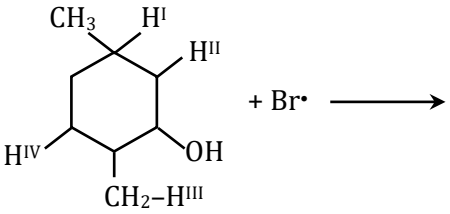
Question	1	2	3	4	5	6	7	8	
Answer	B	C	B	D	C	D	ABCD	A	

For More Join: @IITJEE_Advanced

RACE Regular Analysis Through Continuous Exercise

Reaction Mechanism : RACE-02

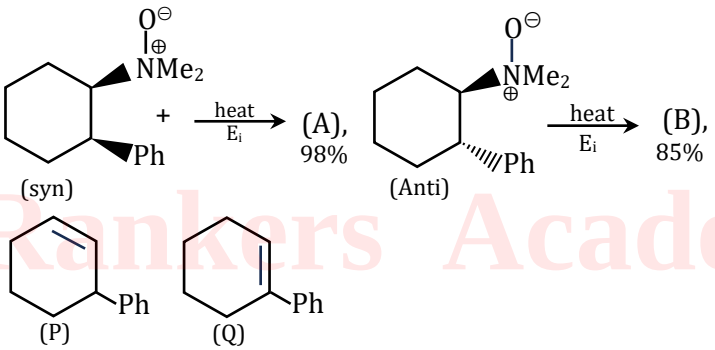
Freeradical substitutaion, Elimination Reactions

1. 

Br˙ will abstract which of the hydrogen most readily ?

(A) I (B) II (C) III (D) IV

CRM140

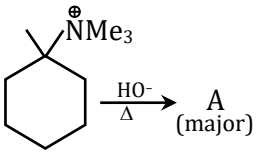
2. 

Rankers Academy JEE

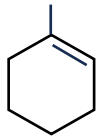
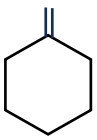
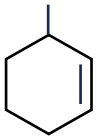
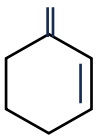
Product (A) and (B) of the above reaction is

(A) A = P, B = P (B) A = Q, B=Q (C) A= P, B= Q (D)A = Q, B= P

CRM141

3. 

Major product is

(A)  (B)  (C)  (D) 

CRM142

4. The conversion of 2,3–dibromobutane to 2–butene with Zn is

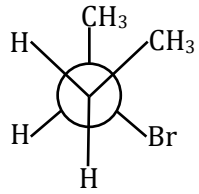
(A) Redox reaction (B) α–Elimination

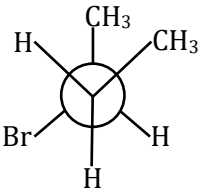
(C) β–Elimination (D) Both α–elimination and –2 butene?

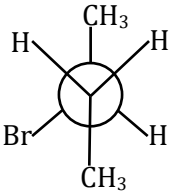
CRM143

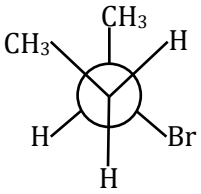
For More Join: @IITJEE_Advanced

5. In the dehydrohalogenation of 2-bromobutane, which conformations leads to the formation of cis-2- butene ?

(A) 

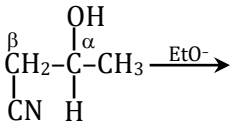
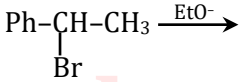
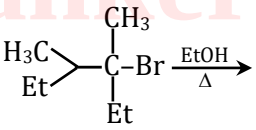
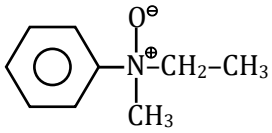
(B) 

(C) 

(D) 

CRM144

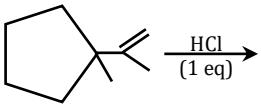
6. Match the items given in column I with those in column II.

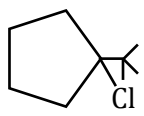
	Column–I		**Column–II**
a.		p.	E1 (elimination unimolecular)
b.		q.	E2 (eliminaiton bi-molecular)
c.		r.	E1 cb (elimination from conjugate base)
d.		s.	E_i (Elimination intramolecular)

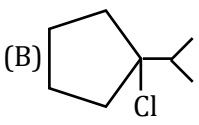
(A) (a→r), (b→q), (c→p), (d→s)
(B) (a→p), (b→q), (c→r), (d→s)
(C) (a→s), (b→r), (c→q), (d→p)
(D) (a→r), (b→q), (c→s), (d→p)

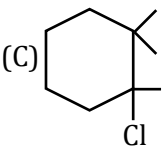
CRM145

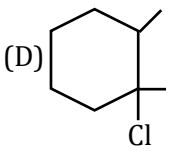
7. Major product of the reaction is



(A) 

(B) 

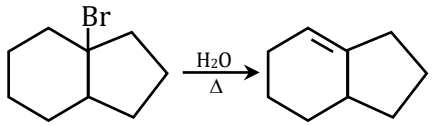
(C) 

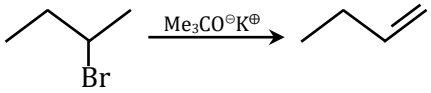
(D) 

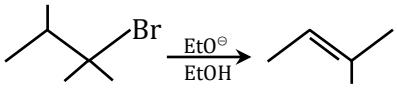
CRM146

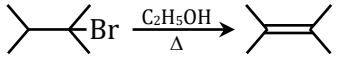
Rankers Academy JEE
For More Join: @IITJEE_Advanced

8. In which of the folloiwng cases, the major product has been correctly shown?

(A) 

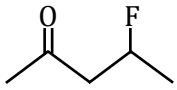
(B) 

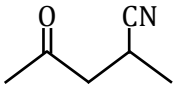
(C) 

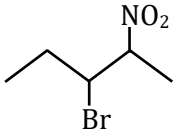
(D) 

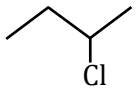
CRM147

9. Which of the following can give E_1 cb reaction?

(A) 

(B) 

(C) 

(D) 

CRM148

Rankers Academy JEE

ANSWER KEY

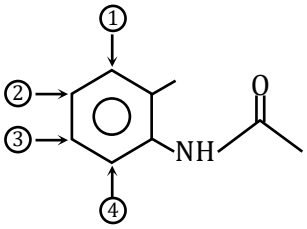
Question	1	2	3	4	5	6	7	8	9	
Answer	A	C	B	C	A	A	C	BCD	AB	

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Reaction Mechanism : RACE-03

ESR, Electrophilic Addition, Nucleophilic Substitution

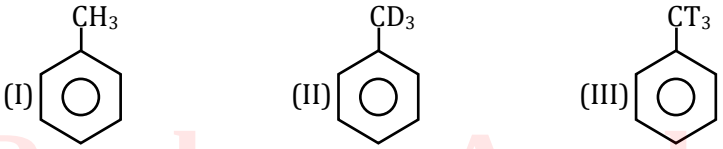
1. 

Sulphonation is the most favourable at the carbon number ...

(A) 1 (B) 2 (C) 3 (D) 4

CRM149

2. Which of the following shows highest reactivity towards EAS (electrophilic aromatic substitution).

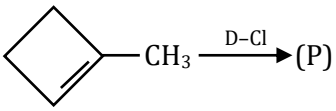


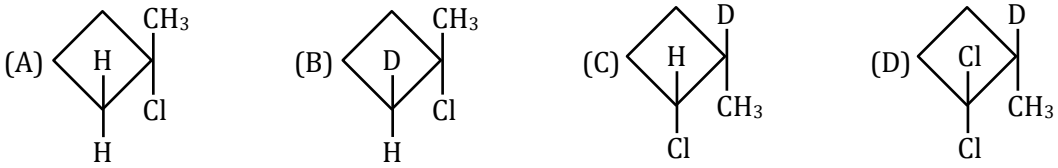
(A) 1 (B) 2 (C) 3 (D) 4

Rankers Academy JEE

CRM150

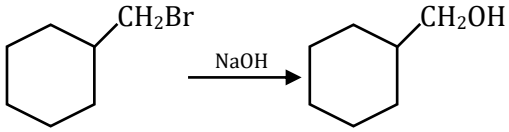
3. Major product (P) in the following raction is





CRM151

4. The suitable solvent for the reaction given below is

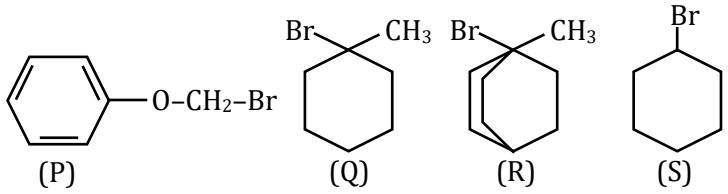


(A) H_2O (B) C_2H_5OH (C) $HCONMe_2$ (D) C_6H_6

CRM152

For More Join: @IITJEE_Advanced

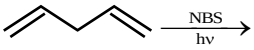
5. Rate of reaction of the given compounds with aqueous ethanol follows the order :



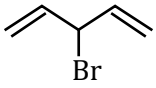
(P) (Q) (R) (S)

(A) P > Q > S > R
(B) Q > P > R > S
(C) P > R > Q > S
(D) R > P > S > Q

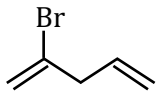
CRM153

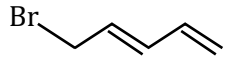
6. Which of the following product will be formed by the reaction 

(P) ?

(A) 

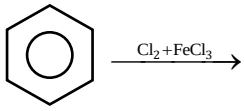
(B) HBr

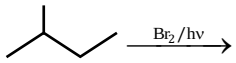
(C) 

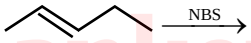
(D) 

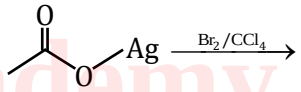
CRM154

7. Which of the following reactions involve free radical as intermediate ?

(A) 

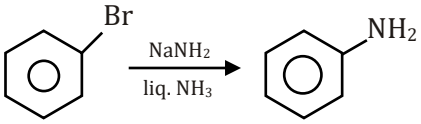
(B) 

(C) 

(D) 

Rankers Academy JEE

CRM155

8. 

The reaction is
(A) ArS_N1
(B) ArS_N2
(C) Nucleophilic substitution via benzyne intermediate
(D) None of these

CRM156

ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	B	A	B	C	A	ABD	BCD	C

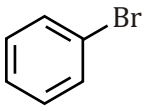
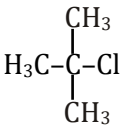
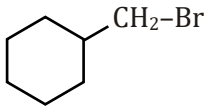
For More Join: @IITJEE_Advanced

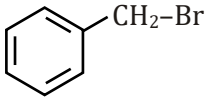
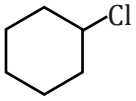
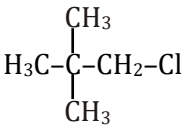
RACE
Regular Analysis Through Continuous Exercise

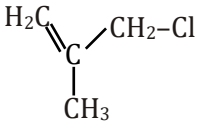
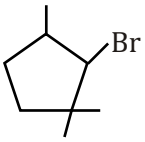
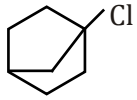
Reaction Mechanism : RACE-04

Nucleophilic Substitution, (SN vs E)

1. Examine the ten structural formulas shown below and select that satisfy each of the following condition. Write one or more (a to j) in each of the question.

(a) 
(b) 
(c) 
(d) CH_3-I

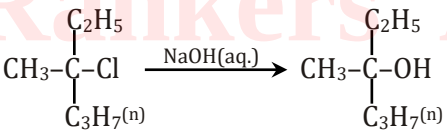
(e) 
(f) 
(g) 

(h) 
(i) 
(j) 

Which compound do not show S_N2 and Elimination reaction ?

CRM157

Rankers Academy JEE

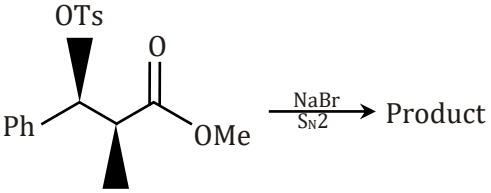
2. 

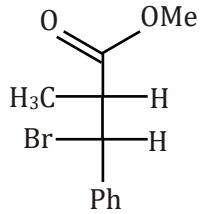
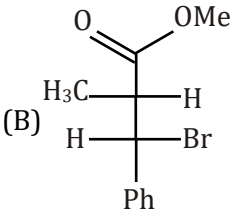
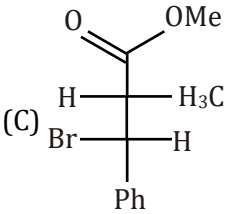
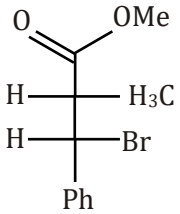
The product obtained will be

(A) *d*- form
(B) *l*- form
(C) racemic mixture mainly
(D) none of these

CRM158

3. What is the product of the following S_N2 reaction?

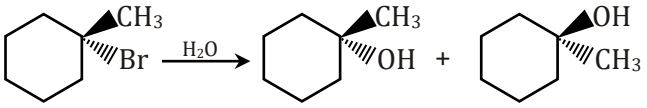


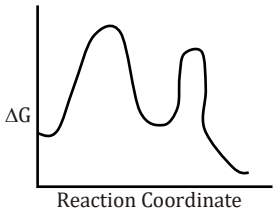
(A) 
(B) 
(C) 
(D) 

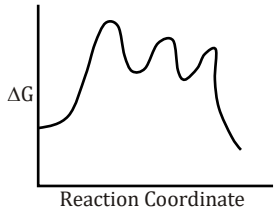
CRM159

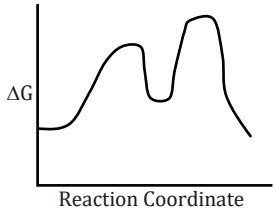
For More Join: @IITJEE_Advanced

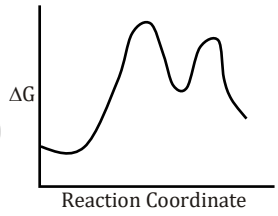
4. Which of the correct reaction coordinate diagram for the following solvolysis reaction?



(A) 

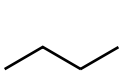
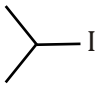
(B) 

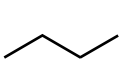
(C) 

(D) 

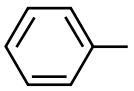
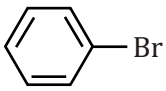
CRM160

5. Choose the correct among the following statements.

(A)  will react more readily than  for S_N2 reaction

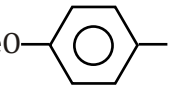
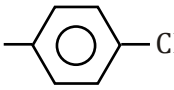
(B)  will react more readily than  for S_N2 reaction

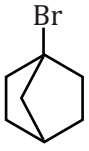
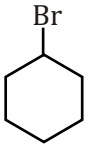
(C)  will react more readily than  for S_N2 reaction

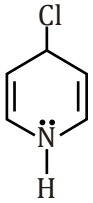
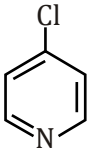
(D)  will react more readily than  for S_N2 reaction.

CRM161

6. Consider the following statement and pick up the correct statement (s).

(A)  will react more readily than  for S_N1 reaction.

(B)  will react more readily than  for S_N1 reaction.

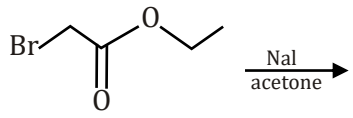
(C)  will react more readily than  for S_N1 reaction.

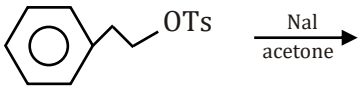
(D) S_N1 reaction occurs in polar protic solvent.

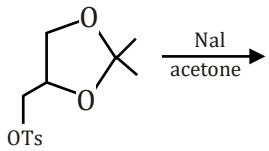
CRM162

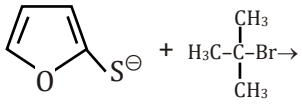
For More Join: @IITJEE_Advanced

7. Which of the following reaction takes(s) place by S_N2 reaction?

(A) 

(B) 

(C) 

(D) 

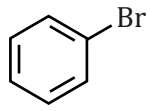
CRM163

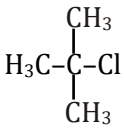
8. Which of the following staments is/are correct about the reactivities of (I) *methyl* chloride and (II) allyl chloride ?

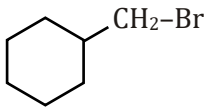
(A) Rate of S_N1 reaction (I) >(II).
(B) Rate of S_N1 reaction (II) >(I).
(C) Rate of S_N2 reaction (I) >(II).
(D) Rate of S_N2 reaction (II) >(I).

CRM164

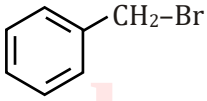
9. Examine the ten structural formulas shown below and select that satisfy each of the following condition. Write one or more (a to j) in each of the question.

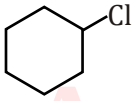
(a) 

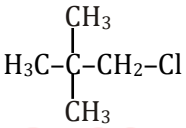
(b) 

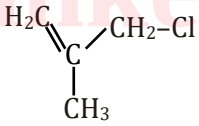
(c) 

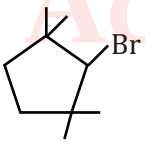
(d) $CH_3–I$

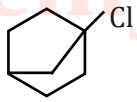
(e) 

(f) 

(g) 

(h) 

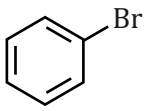
(i) 

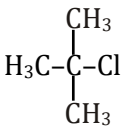
(j) 

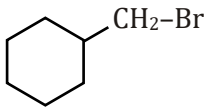
Which compounds give an S_N2 substitution reaction on treatment with alcoholic NaSH?

CRM165

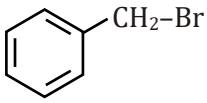
10. Examine the ten structural formulas shown below and select that satisfy each of the following condition. Write one or more (a to j) in each of the question.

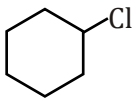
(a) 

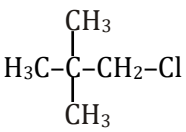
(b) 

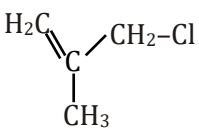
(c) 

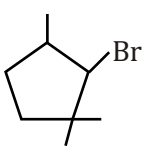
(d) $CH_3–I$

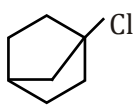
(e) 

(f) 

(g) 

(h) 

(i) 

(j) 

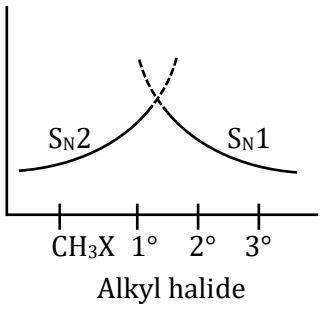
Which compounds give an E2 elimination reaction on treatment with alcoholic KOH?

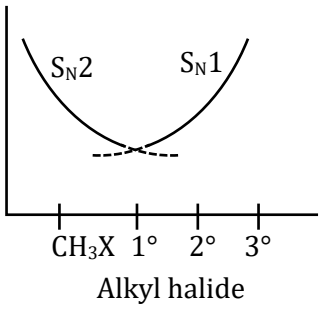
CRM166

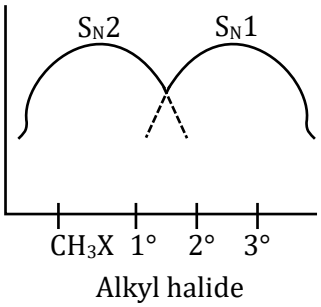
Rankers Academy JEE

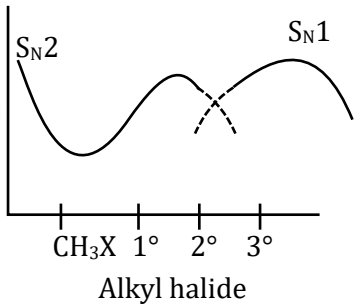
For More Join: @IITJEE_Advanced

11. Which of the following curves correctly represents S_N1 vs S_N2 ?

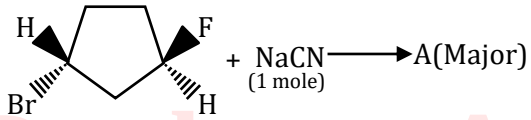
(A) 

(B) 

(C) 

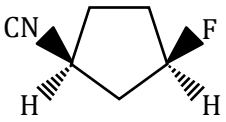
(D) 

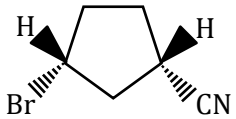
CRM167

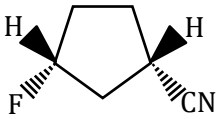
12. 

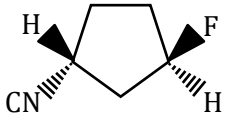
Rankers Academy JEE

The major product of this reaction is/are

(A) 

(B) 

(C) 

(D) 

CRM168

13. The products of reaction of alcoholic silver nitrite with ethyl bromide are

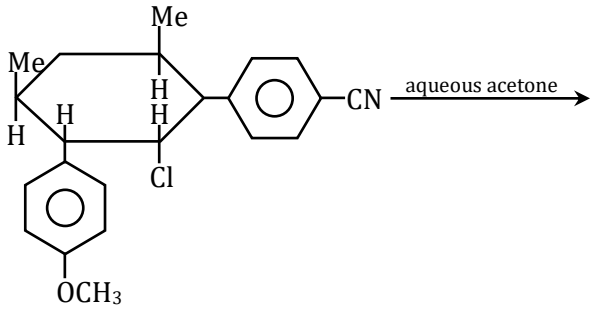
(A) Ethyne (B) Ethene

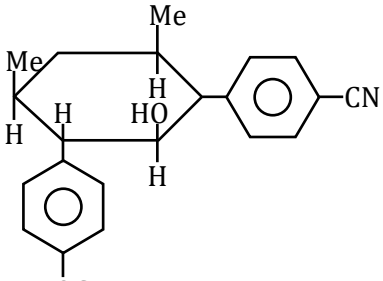
(C) Nitroethane (D) Ethyl nitrite

CRM169

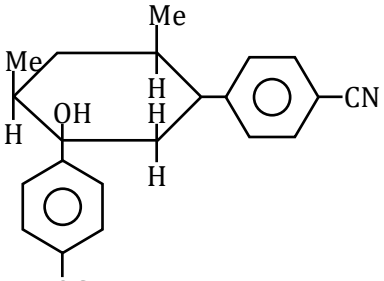
For More Join: @IITJEE_Advanced

14. Which of the following are correct regarding the products of the given reaction?

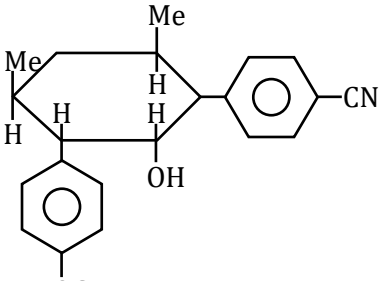




(X)



(Y)



(Z)

(A) only X and Z are formed
(B) Y > Z (amount)
(C) X > Y (amount)
(D) Y > X (amount)

CRM170

Rankers Academy JEE

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	3	C	A	A	ACD	ACD	ABC	BC	5	4
Question	11	12	13	14						
Answer	B	AC	CD	BD						

For More Join: @IITJEE_Advanced